

Daniel Mikhail

Software Engineer

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Software engineer with 6+ years across platform services, data pipelines, and full-stack systems. I build internal tooling and Go service patterns that make platform work tractable — most recently on healthcare platform infrastructure at Verily.

EXPERIENCE

Verily Life Sciences (Alphabet), Software Engineer | (Boston, MA)

July 2022 - Present

Cross-System Technical Leadership

- **Lead engineer for Disenrollment** — gathered and reconciled requirements from clinical experts, upstream UX/PM, and 4+ senior/staff engineers across adjacent teams, translating them into the core implementation and the data contracts between them
- **Made the call on CloseAccount's implementation pattern** — single backend request triggering distributed side effects rather than frontend fanout; scaffolded for 3-engineer parallel development with a code path reusable for backfills
- **Co-Lead of the team's mentorship program**, coordinating ~20 active mentor/mentee pairings, designing lesson plans and program length

Platform Tooling & Developer Experience

- Designed a reusable backfill framework with injection points for provenance, scoped rate limiting (only on cascading non-provenance writes), and pagination/logging — reduces dev time and error surface; runs 2–3 backfills/month, ~1,000 patients per run, **emergency rollouts in under 3 days**
- Built a multithreaded FHIR wrapper handling rate limiting, safe pagination, and core resource extraction from bundles — enables **high-confidence backfills** against queries with unpredictable size and rate constraints
- Authored comprehensive developer guides and contract tests establishing best practices for service integration

Participant System

- Designed the **“participant attribute” pattern** after surfacing that downstream teams kept needing data shapes the core model couldn't serve — a foreign-key layer (FHIR Basic resources) that lets each consumer request what they need without core-model changes; decoupled cross-team velocity from data-model decisions
- Led refactoring of a core data model and system, improving maintainability through service modularization and third-party integration abstraction
- Reduced testing cycle time by **75%** through decoupling frontend/backend test suites in Consent System

Provider Console

- Established frontend architecture patterns by implementing core React/TypeScript components, creating a flagship MFE that became **standard reference across 4+ teams**
- Broke down card implementation into modular Backend-for-Frontend (BFF) architecture enabling parallel development across junior engineers, reducing MVP delivery time by **50%**

Verizon Media Group (Yahoo!), Software Production Engineer | (Champaign, IL)

August 2018 - March 2021

Production point of contact for data pipelines and tools on the on-prem Hadoop ecosystem (Pig, Oozie, Spark) on AWS and RedHat clusters.

- Developed a full-stack monitoring tool for multiple Lambda-architecture streaming pipelines, providing real-time pipeline health visibility
- Escalated urgent feature requests across the pipeline, like new security requirements (COPPA, GDPR, etc.), deployed ahead of schedule; coordinated Spectre/Meltdown patching across on-prem RedHat clusters
- Formed a high-level intuition of the pipeline to catch potential SLA misses early, and pass on tribal knowledge to newer Production Engineers
- Hands-on with key and group management, replication, load balancing, networking, CI/CD, and Linux Bash scripting

SKILLS

Languages & infra: Go, Python, TypeScript/React, Java · Docker/Kubernetes, GCP, AWS, CI/CD

Development: API Design, Distributed Systems, Developer Tooling, Dependency Injection · **Domains:** Platform/Infra, FHIR

EDUCATION

- **Masters of Computer Science**, University of Illinois at Urbana-Champaign — December 2024 (finished while at Verily)
- **BSc, Computer Science**, Rose-Hulman Institute of Technology (Terre Haute, IN) — May 2018